

Christoffel Symbols via Covariant Derivative (Polar Coordinates)

1. Coordinate System

We use planar polar coordinates:

$$(r, \theta)$$

Position vector:

$$\mathbf{r} = r \cos \theta \hat{\mathbf{x}} + r \sin \theta \hat{\mathbf{y}}$$

2. Coordinate Basis

$$\mathbf{e}_r = \frac{\partial \mathbf{r}}{\partial r} = \cos \theta \hat{\mathbf{x}} + \sin \theta \hat{\mathbf{y}}$$

$$\mathbf{e}_\theta = \frac{\partial \mathbf{r}}{\partial \theta} = -r \sin \theta \hat{\mathbf{x}} + r \cos \theta \hat{\mathbf{y}}$$

3. Metric

$$g_{ij} = \mathbf{e}_i \cdot \mathbf{e}_j$$

$$g_{rr} = 1, \quad g_{\theta\theta} = r^2, \quad g_{r\theta} = 0$$

4. Dual Basis

The dual basis satisfies:

$$\mathbf{e}^i \cdot \mathbf{e}_j = \delta_j^i$$

Thus:

$$\mathbf{e}^r = \mathbf{e}_r$$

$$\mathbf{e}^\theta = \frac{1}{r^2} \mathbf{e}_\theta$$

5. Key Formula

$$\Gamma_{ij}^k = \mathbf{e}^k \cdot \frac{\partial \mathbf{e}_i}{\partial x^j}$$

We compute all derivatives explicitly.

6. Derivatives of Basis Vectors

(a) $\frac{\partial \mathbf{e}_r}{\partial r}$

$$\frac{\partial \mathbf{e}_r}{\partial r} = 0$$

Thus:

$$\Gamma_{rr}^r = \mathbf{e}^r \cdot 0 = 0 \quad \Gamma_{rr}^\theta = \mathbf{e}^\theta \cdot 0 = 0$$

(b) $\frac{\partial \mathbf{e}_r}{\partial \theta}$

$$\frac{\partial \mathbf{e}_r}{\partial \theta} = -\sin \theta \hat{\mathbf{x}} + \cos \theta \hat{\mathbf{y}} = \frac{1}{r} \mathbf{e}_\theta$$

Projection:

$$\Gamma_{r\theta}^r = \mathbf{e}^r \cdot \frac{1}{r} \mathbf{e}_\theta = 0$$

$$\Gamma_{r\theta}^\theta = \mathbf{e}^\theta \cdot \frac{1}{r} \mathbf{e}_\theta = \frac{1}{r^2} \mathbf{e}_\theta \cdot \frac{1}{r} \mathbf{e}_\theta = \frac{1}{r^2} \cdot \frac{1}{r} \cdot r^2 = \frac{1}{r}$$

(c) $\frac{\partial \mathbf{e}_\theta}{\partial r}$

$$\frac{\partial \mathbf{e}_\theta}{\partial r} = -\sin \theta \hat{\mathbf{x}} + \cos \theta \hat{\mathbf{y}} = \frac{1}{r} \mathbf{e}_\theta$$

Projection:

$$\Gamma_{\theta r}^r = \mathbf{e}^r \cdot \frac{1}{r} \mathbf{e}_\theta = 0$$

$$\Gamma_{\theta r}^\theta = \mathbf{e}^\theta \cdot \frac{1}{r} \mathbf{e}_\theta = \frac{1}{r}$$

(d) $\frac{\partial \mathbf{e}_\theta}{\partial \theta}$

$$\frac{\partial \mathbf{e}_\theta}{\partial \theta} = -r \cos \theta \hat{\mathbf{x}} - r \sin \theta \hat{\mathbf{y}} = -r \mathbf{e}_r$$

Projection:

$$\Gamma_{\theta\theta}^r = \mathbf{e}^r \cdot (-r \mathbf{e}_r) = -r$$

$$\Gamma_{\theta\theta}^\theta = \mathbf{e}^\theta \cdot (-r \mathbf{e}_r) = 0$$

7. Final Result

$$\Gamma_{\theta\theta}^r = -r$$

$$\Gamma_{r\theta}^\theta = \Gamma_{\theta r}^\theta = \frac{1}{r}$$

All other Christoffel symbols vanish.